Section I: Multiple Choice [50]

1. [1] When a light ray passes from a medium of lower refractive index to a medium of higher refractive index, the ray bends ______ the surface normal.
   a. Towards
   b. Away from
   c. Neither (the ray does not bend)

2. [2] Brewster's angle is defined as the incidence angle at which unpolarized light is fully polarized when it is reflected from a smooth surface. It can be mathematically computed by the following relation:

$$tan(\theta_p) = \frac{n_2}{n_1}$$

   where θ_p is Brewster's angle. n_2 and n_1 are the refractive indices of the media the light passes into and in from, respectively. Knowing that glass has a refractive index of 1.52, what is Brewster's angle for a light ray passing from a vacuum into a glass window?
   a. 90°
   b. 74°
   c. 57°
   d. 33°

3. [1] Consider the light ray from the previous question. Assuming it hits the window at Brewster's angle, the angle between the reflected and refracted ray will be ______.
   a. 90°
   b. 74°
   c. 57°
   d. 33°

4. [1] Which of the following is an appropriate measure of radiometric resolution?
   a. Image scale
   b. Ground sampling distance
   c. Number of bands
   d. Bits

5. [3] In LiDAR systems, rotation matrices are used to translate between coordinate systems. A 3D rotation matrix can be defined by three parameters, namely the roll (ω), pitch (ϕ), and yaw (κ). These angles indicate the rotation about the x, y, and z axes (respectively) of the sensor relative to the ground. Imagine you have a vector in the ground coordinate system $[x_{\omega\phi\kappa};\ y_{\omega\phi\kappa};\ z_{\omega\phi\kappa}]$ that you wish to transform into the sensor coordinate system (i.e. $[x;\ y;\ z]$). If the rotation matrix is $R^{XYZ}_{X_{\omega\phi\kappa}Y_{\omega\phi\kappa}Z_{\omega\phi\kappa}}$ and $[x;\ y;\ z] = R^{XYZ}_{X_{\omega\phi\kappa}Y_{\omega\phi\kappa}Z_{\omega\phi\kappa}}[x_{\omega\phi\kappa};\ y_{\omega\phi\kappa};\ z_{\omega\phi\kappa}]$, which of the following rotation matrices is correct for a 90° roll, 0° pitch, and 0° yaw? *Hint: Use the right hand rule to determine directions of positive rotation. Note that semicolons indicate the start of a new row in the matrix.*
   a. $R^{XYZ}_{X_{\omega\phi\kappa}Y_{\omega\phi\kappa}Z_{\omega\phi\kappa}} = [1\ 0\ 0;\ 0\ 1\ 0;\ 0\ 0\ 1]$
   b. $R^{XYZ}_{X_{\omega\phi\kappa}Y_{\omega\phi\kappa}Z_{\omega\phi\kappa}} = [1\ 0\ 0;\ 0\ 0\ 1;\ 0\ 1\ 0]$

c. $R^{XYZ}_{X_{\omega\phi\kappa}Y_{\omega\phi\kappa}Z_{\omega\phi\kappa}} = [1\ 0\ 0;\ 0\ 0\ -1;\ 0\ 1\ 0]$
d. $R^{XYZ}_{X_{\omega\phi\kappa}Y_{\omega\phi\kappa}Z_{\omega\phi\kappa}} = [1\ 0\ 0;\ 0\ 0\ 1;\ 0\ -1\ 0]$

6. [2] Use the rotation matrix from the previous question. If $[x_{\omega\phi\kappa};\ y_{\omega\phi\kappa};\ z_{\omega\phi\kappa}] = [1;\ 2;\ 3]$, what is $[x;\ y;\ z]$?
   a. $[x;\ y;\ z] = [1;\ 2;\ 3]$
   b. $[x;\ y;\ z] = [3;\ 2;\ 1]$
   c. $[x;\ y;\ z] = [1;\ 3;\ 2]$
   d. $[x;\ y;\ z] = [1;\ -3;\ 2]$

7. [1] Which of the following statements about the electromagnetic spectrum is true?
   a. Microwaves are higher in energy than ultraviolet rays.
   b. Compared with other types of electromagnetic radiation, radio waves have a low frequency.
   c. Yellow light has a shorter wavelength than blue light.
   d. Skin cancer is primarily caused by the Sun's infrared radiation.

8. [1] As an object's temperature increases, will its blackbody curve shift towards shorter or longer wavelengths? Which law is this given by?
   a. Shorter; Stefan-Boltzmann's Law
   b. Longer; Stefan-Boltzmann's Law
   c. Shorter; Wien's Law
   d. Longer; Wien's Law

9. [1] Radar range resolution is given by the following equation:

$$\rho = \frac{c\tau}{2}$$

   where ρ is the range resolution, c is the speed of light, and τ is the pulse duration. If you wish to achieve a range resolution of 1 cm (the range resolution of sea surface altimetry is typically around this value), how long does your radar pulse need to be?
   a. 3.33e-10 s
   b. 3.33e-11 s
   c. 6.67e-10 s
   d. 6.67e-11 s
10. [2] What is the signal bandwidth based on this pulse duration?
   $(B = \frac{1}{\rho})$
   a. 3e10 Hz
   b. 3e11 Hz
   c. 1.5e10 Hz
   d. 1.5e11 Hz
11. [1] Which of the following best describes relief displacement distortion in vertical aerial photography?
   a. Image features away from the nadir are compressed due to the rotating motion of the sensor
   b. Objects at the nadir will only have their tops visible, while other objects appear to lean away from the nadir
   c. Each successive scan line of a satellite is slightly west of the previous due to the rotation of the Earth
   d. Straight lines appear curved due to the spherical shape of the lens

12. [1] Geostationary orbits, such as that followed by NASA's GOES satellite network, are approximately ______ above the ______.
    a. 36,000 km; Equator
    b. 36,000 km; poles
    c. 24,000 km; Equator
    d. 24,000 km; poles
13. [1] The natural, cyclic climate variation due to changes in the Earth's orbit and axial tilt is commonly referred to as which of the following?
    a. Keeling cycles
    b. Milankovitch cycles
    c. El Nino
    d. Global warming
14. [3] Which of the following statements about satellite gravimetry is true? Select all that apply.
    a. It is desirable for the satellite to be in a high-earth orbit.
    b. Sea level rise and ice mass loss are two major climate change-related applications of satellite gravimetry.
    c. The Gravity Recovery and Climate Experiment (GRACE) satellites, first launched in 2002, are still in orbit.
    d. The two modes for satellite-to-satellite tracking are high-low (SST-HL) and low-low (SST-LL).
    e. The GRACE Follow-On mission is capable of measuring the distance between twin satellites to nanometer accuracy using laser interferometry.
    f. The GRACE satellites are/were in the same altitude orbit.
15. [3] The radar backscatter coefficient β^0 is influenced by which of the following? Select all that apply.
    a. Terrain features
    b. Radar echo power
    c. Incidence angle
16. [1] Which of the following is not an important scattering effect in polarimetric SAR?
    a. Volume scattering
    b. Surface/specular scattering
    c. Triple-bounce backscatter
    d. Double-bounce backscatter
17. [2] Which of the following statements about polarimetric SAR is true? Select all that apply.
    a. Quad-pol systems are capable of producing VV, HH, HV, and VH imagery.
    b. Dual-pol systems can produce HV or VH imagery, but not VV or HH.
    c. A smooth surface, such as water, produces co-polarized (VV or HH) backscatter.
    d. Helix scattering is most likely to occur in urban areas with tall buildings.
18. [2] The global warming potentials (GWPs) of greenhouse gases are calculated in reference to which gas?
    a. SO2
    b. H2O
    c. CO2
    d. CH4
19. [2] Which of the following is the main source of atmospheric SF6?
    a. Vehicular emissions
    b. Volcanic eruptions
    c. Fossil fuel burning
    d. Power appliance leakage
20. [1] Chlorine in CFCs reacts with atmospheric ozone to produce which immediate products?
    a. ClO and O

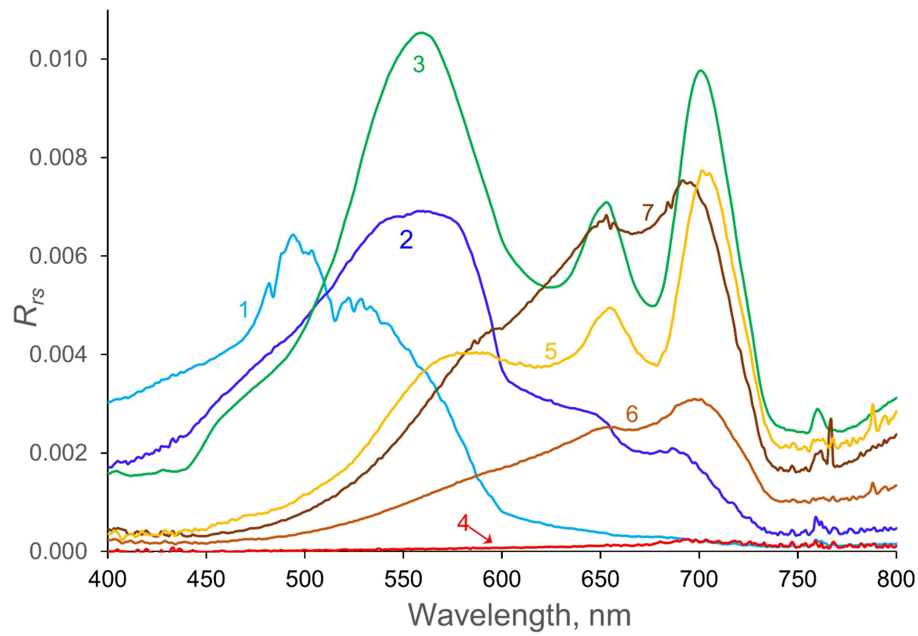
b. ClO and O2
c. Cl and O
d. Cl and O2

21. [1] Which of the following is not an advantage of using indirect georeferencing (i.e. ground control points) over direct georeferencing (GNSS/INS units)?
    a. Higher accuracy
    b. Less time spent in the field
    c. Ability to correct for sensor and terrain distortions in post-processing
    d. Improved versatility with various types of drones and sensors
22. [3] NSS includes which of the following satellite systems? Select all that apply.
    a. GOES
    b. GLONASS
    c. GPS
    d. Galileo
    e. JPSS
    f. BeiDou
23. [3] LandSat's OLI (Operational Land Imager) operates in which regions of the electromagnetic spectrum? Select all that apply.
    a. Radio
    b. Microwave
    c. Near infrared
    d. Shortwave infrared
    e. Visible
    f. UV
24. [1] The Advanced Baseline Imager (ABI) aboard GOES-R satellites is capable of measuring radiation in how many spectral bands?
    a. 12
    b. 14
    c. 16
    d. 18
25. [1] Of ABI's bands, how many are in the near-infrared region?
    a. 2
    b. 4
    c. 6
    d. 8
26. [1] Landsat 8 and 9's Operational Land Imager (OLI) contain a panchromatic band (Band 8). What is the purpose of this band?
    a. Data collection in variable weather conditions
    b. Calculation of NDVI
    c. Imaging clouds at the top of the atmosphere
    d. Improving spatial resolution of color images through pan-sharpening
27. [3] MODIS data products are produced in which of the following resolutions? Select all that apply.
    a. 50 m
    b. 100 m
    c. 250 m
    d. 500 m

    e. 1 km
    f. 10 km
28. [1] NASA's Terra and Aqua satellites, which carry MODIS, are flying in which type of orbit?
    a. Geostationary
    b. Sun-synchronous
    c. Geosynchronous
    d. Geostationary transfer
29. [1] MODIS bands 1 - 19 and 26 are unique from its other bands in which of the following ways
    a. These bands measure reflected sunlight.
    b. These bands are exclusively shortwave-infrared.
    c. These bands are exclusively in near-infrared.
    d. These bands are used for cloud detection and monitoring.
30. [1] MODIS bands 20 - 25 and 27 - 36 are unique from its other bands in which of the following ways?
    a. These bands only operate during the day.
    b. These bands only operate at night.
    c. These bands primarily measure thermal emissivity from clouds, the atmosphere, and the Earth's surface.
    d. These bands primarily measure reflected sunlight.
31. [2] Bowtie artifact is generated at the ______ of images due to ______.
    a. Middle; atmospheric distortion
    b. Middle; varying ground sampling distance with scan angle
    c. Edge; atmospheric distortion
    d. Edge; varying ground sampling distance with scan angle

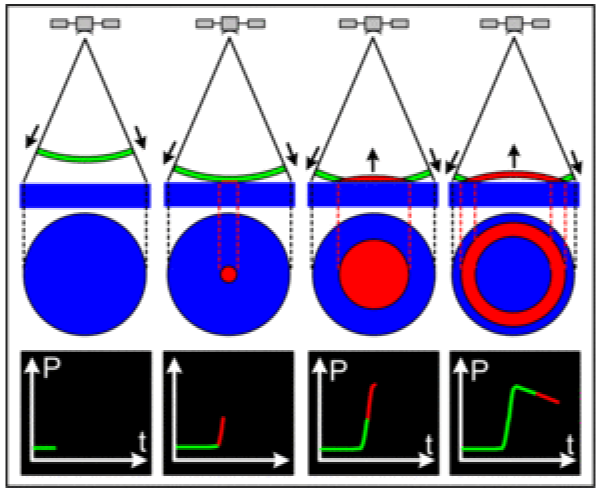
Short Answer [79]

32. [11] The following graph shows the reflectance spectra (the fraction of incident light that is reflected) for several lakes in Minnesota (numbered 1 - 7).

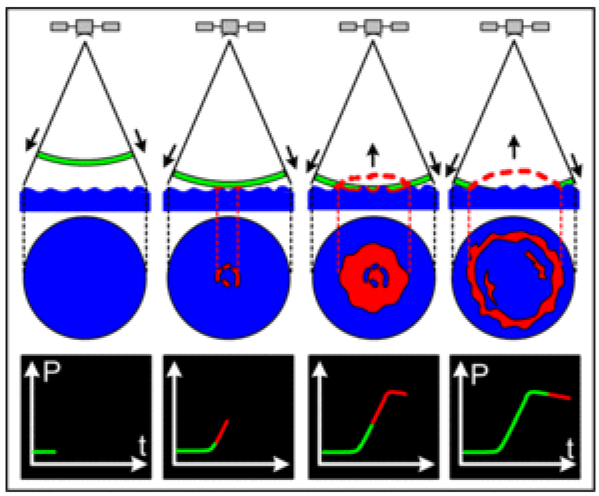


   a. [1] One of these lakes was formed from an abandoned mine pit. Consequently, it is very deep and clear with low nutrients. Which lake (1 - 7) is this?
   b. [3] Curves 4, 5, 6, and 7 all show low reflectance in the 400 - 500 nm range. What does this suggest about the organic matter content of these lakes?
   c. [2] What is the formula for the Normalized Difference Water Index (NDWI)?
   d. [3] Imagine you have images of Lakes 1, 3, and 7, select a representative pixel from each, and calculate the NDWI. Rank the lakes from high to low in terms of expected NDWI. You do not need to provide any numerical estimates.
   e. [2] Why is NDWI a better index to use in this case than NDVI?

33. [7] Sea surface altimetry is a type of radar used for measuring the height of the sea surface. The shape of the *return waveform* (graph of the intensity of the reflected wave over time) can be used to estimate sea surface parameters.
   a. [1] Which of the following missions did NOT measure sea surface altimetry?
      i. TOPEX/Poseidon
      ii. Jason-3
      iii. Landsat
      iv. Sentinel-6
   b. [4] The diagram below shows the idealized shape of the return waveform over a perfectly smooth ocean.

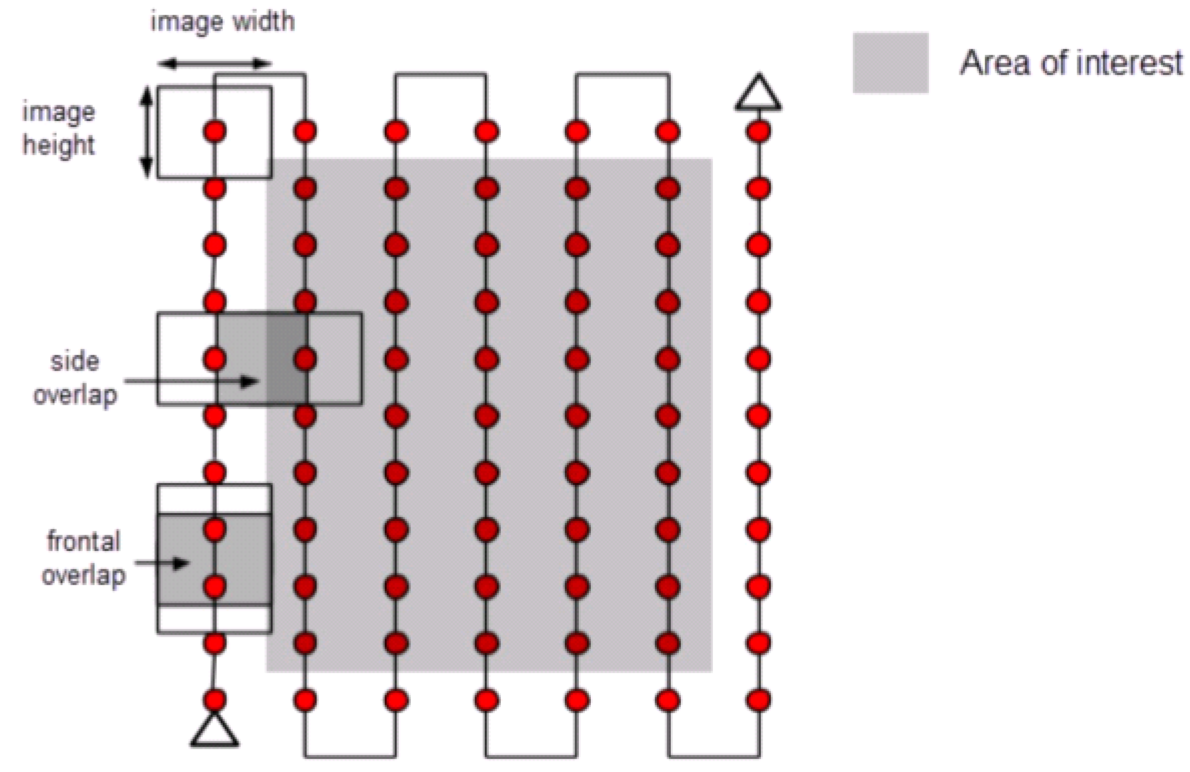


Observe that the intensity of the return waveform increases sharply as the radar pulse hits the smooth ocean surface. If the ocean was choppy with lots of waves, how would the shape of the return waveform look different? Why?



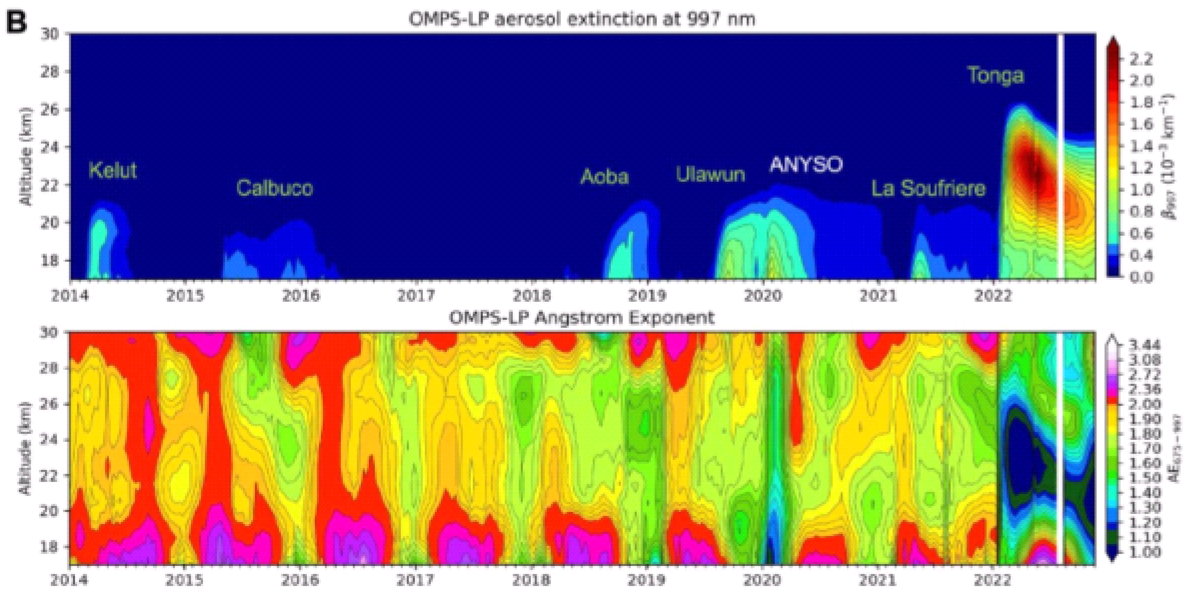
c. [2] What sea surface parameter is being measured in part (b)?

34. [3] Synthetic aperture radar (SAR) is a specialized type of radar that is increasingly being used in oceanographic research.
    a. [1] SAR is superior to traditional radar in what type of resolution?
    b. [2] Both spaceborne and airborne SAR systems are used in oceanographic applications. Which of the following is NOT an advantage of airborne SAR over spaceborne SAR? Select all that apply.
        i. Airborne SAR can generally achieve wider swaths with a smaller range of incident angles.
        ii. Airborne SAR is more flexible in terms of the revisit period.
        iii. Airborne SAR images are less susceptible to geometric distortion.
        iv. Airborne SAR can easily cover a larger area than spaceborne SAR.
35. [9] When planning a mission to map an area, it is important to ensure that the entire area is covered in the most efficient method possible. You are in the process of designing a mission to image a patch of recently burned forest. A diagram showing a concept of the proposed mission is shown below, where each red dot represents an image and each north/south row of images represents a flight line.



a. [3] If the target area has significant variation in terrain (such as an abundance of hills), how will the frontal overlap be affected? Why? Assume that the distance travelled by the drone between successive images is the same.
b. [3] What effect will your answer to the previous question have on the quality of data gathered by the mission? Hint: the desired overlap is typically 70-80%. Consider the purpose of overlap in image processing.
c. [1] What can you do to avoid this issue?
d. [2] List two issues associated with having a frontal overlap value significantly above 80%.

36. [2] How much time does it take for a geostationary satellite to complete one full orbit? What advantage does this provide the satellite for research purposes?
37. [16] The figure below shows 10-day mean measurements of β997 and AE675-997 taken by the Ozone Mapping and Profiling Suite Limb Profiler (OMPS-LP), averaged over 20˚N to 30˚S, between 2014 and 2022. The anomalies shown in the top profile (Kelut, Calbuco, etc.) correspond to volcanic eruptions.



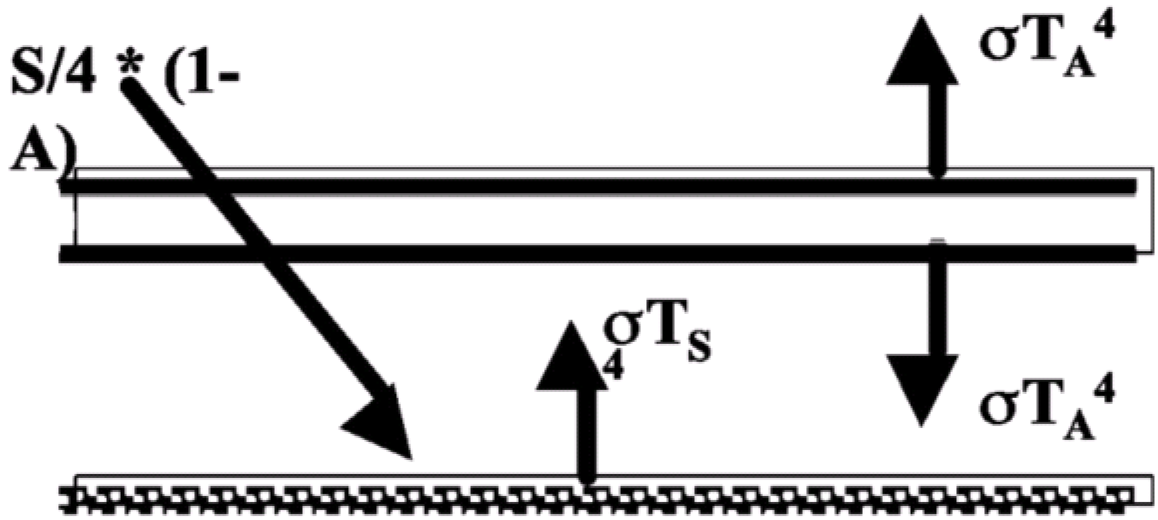
a. [2] Which two satellites currently carry OMPS-LP?
b. [2] This diagram gives profiles for two different variables: the aerosol extinction coefficient at 997 nm (top), and the Angstrom exponent at 675-997 nm (bottom). What aerosol properties do these two values describe, respectively?
c. [3] OMPS-LP measures the aerosol extinction coefficient at six different wavelengths: 510, 600, 675, 745, 869, and 997 nm. What region of the electromagnetic spectrum are these wavelengths in? What is the purpose of measuring the coefficient at six different wavelengths rather than just one?
d. [3] Note that in the bottom profile, there is a yearly occurrence (roughly in the summer) of high values of the Angstrom exponent near the surface. What does this indicate about the aerosol layer? What is a possible explanation for this phenomenon?
e. [4] The Tonga event in 2022 released large amounts of water vapor into the air, leading to the significant anomaly visible in the profiles. How does water vapor in volcanic eruptions affect aerosol growth?
f. [2] What climate forcing effects to volcanic eruptions have, considering only aerosol formation?

38. [7]
    a. [2] The Sun's temperature is around 6100 K. What is its wavelength of peak emission? Indicate the formula you used and round your answer to 3 significant figures.
    b. [2] If the Sun is a perfect blackbody, what is its radiated power in W/sq. m? Indicate the formula you used and round your answer to 3 significant figures.
    c. [3] Is the assumption of blackbody radiation in part (b) realistic? Why?

39. [11] The information below will help you to answer the next few questions. Show critical steps in your calculations.
    Solar luminosity (Q): $3.9*10^{26}$ W
    Distance between Sun and Earth (D): $1.5*10^{8}$ km
    Earth's radius (d): 6,371 km
    a. [3] Calculate the solar energy density at the surface of a sphere with a radius of D. For the purposes of this calculation, assume that the Sun is a point in the center of the sphere.
    b. [4] The value you calculated in the previous question is called the solar constant, and can be denoted with the variable S. Using this value, calculate the expected average temperature of the Earth. Assume that the Earth is a perfect blackbody with no atmosphere. Hint: consider how the Sun and Earth are oriented relative to each other. Does the entire Earth absorb the same amount of radiation?

c. [4] What is the Earth's average albedo? Using this value, recalculate the expected average temperature of the Earth. In this scenario, is the Earth still a blackbody?

40. [13] Now, consider a simple greenhouse effect created by a thin, single-layer atmosphere with an emissivity of 1 (see the diagram below). Note that T_A denotes the temperature at the top of the atmosphere, while T_S is the temperature at the Earth's surface.



a. [3] Write the radiative balance equation for the top of the atmosphere using S (refer to part (b) of the previous question), A (the Earth's albedo), T_A, and the relevant physical constants. You may denote T_A as T_A and Greek letters by spelling them out.

b. [4] Write the radiative balance equation for the bottom of the atmosphere using S, A, T_A, T_S, and the relevant physical constants.

c. [4] Write the overall radiative balance equation for the atmosphere. Using this equation, derive an expression for T_S in terms of T_A.

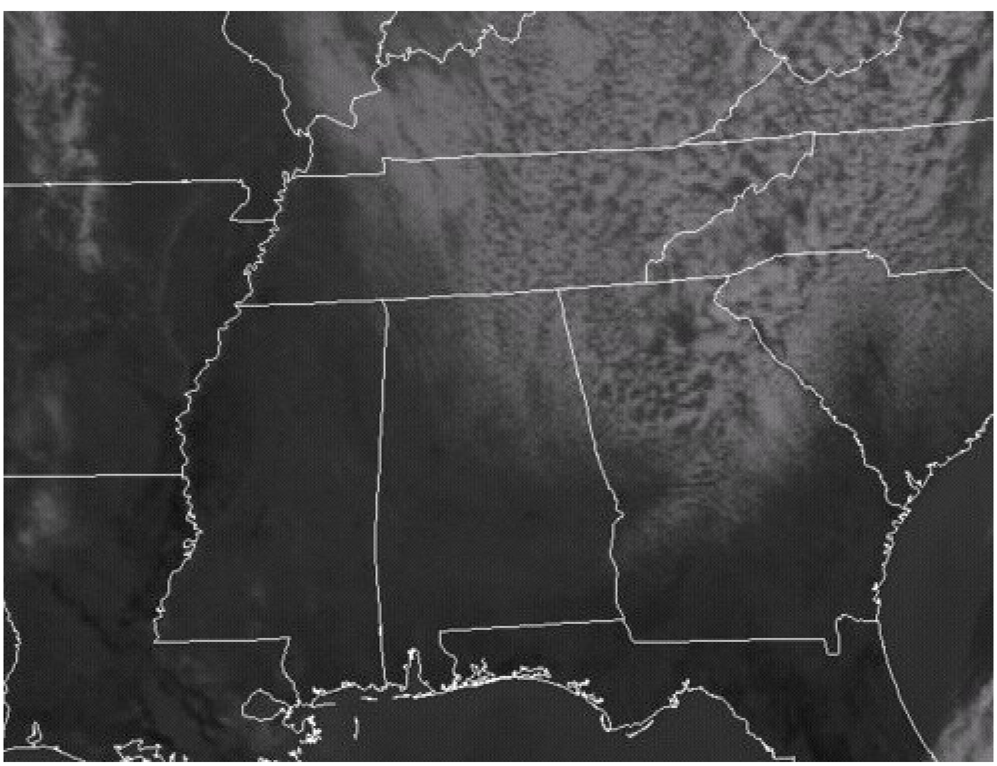
d. [2] Using your calculated T_A from part (c) of the previous question, what is the numerical value of T_S?

Case Studies [131]

Mini Case Study I: Miscalleoneous Mashup [29]

You are currently trapped under the regime of [former disgraced Dr.] Gerald, who has made appearances before in Rickards in years past. This time, he has wormed his way into Remote Sensing, and he has decided to ruin yet ANOTHER field. Bewarned, this time he's gone absolutely broke and thus has nothing to lose. As if his morals weren't already dead and lying in a ditch.

41. [2] Typically, we see 30m resolution as the typical level of spatial resolution seen in most remote sensing instruments. Why?
42. [2] Assume that two giant oaks (*Q. rubra* and *Q. palustris* – try to figure out what type of) shown on a photograph can be located on a 1:15,000 scale topographic map. The measured distance between them is 33.4 mm on the map and 65.5 mm on the photograph.
    a. [1] What is the scale of the photograph?
    b. [1] At that scale, what is the length of a road that measures 49.7 mm on the photograph?
43. [4] State Snell's Law in your own words. To the best of your ability, explain one reason as to why it can be applicable/important in Remote Sensing.
44. [4] Assume a vertical photograph was taken at a flying height of 7500 m above sea level using a camera with a 152-mm-focal-length lens.
    a. [2] Determine the photo scale at points A and B, which lie at elevations of 4200 and 3560 m.
    b. [2] What ground distance corresponds to a 26.8-mm photo distance measured at each of these elevations?
45. [6] Define and differentiate between orthographic and perspective projections. What sort of distortions arise in a map compared to a vertical photo and why?
46. [2] The following shows a clouds over the US, taken as an infrared image:



Source: Theweatherprediction

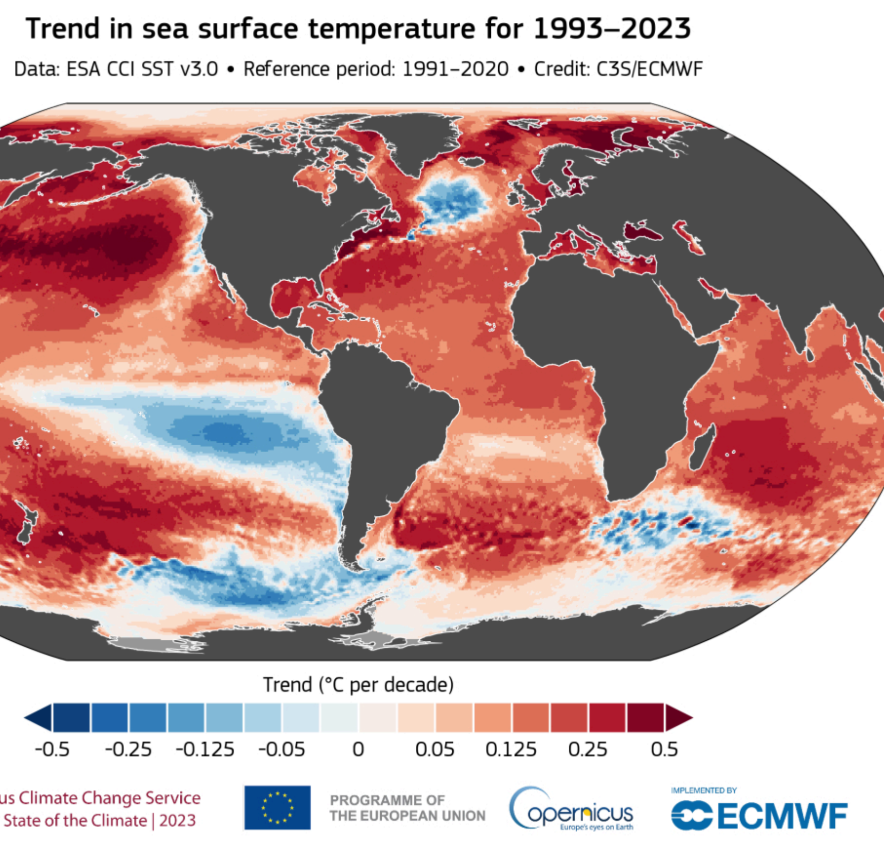
    a. [1] What type of cloud(s) are these depicted above?

b. [1] What does the white represent?

47. [2] Say you want to best show the effects to which a post-burn site of a grassland has endured a few weeks after the initial fire (which was very intense). Would you use true-color or false-color remote sensing? Why?
48. [2] An area which you presume to be a large body of water on a remote sensing image is entirely black on such an image. What imagery was this taken in regards to? Why might bodies of water appear black in this type of imagery?
49. [2] What is a relief displacement?
50. [3] The length of line AB and the elevation of its endpoints, A and B, are to be determined from a stereopair containing images a and b. The camera used to take the photographs has a 165.4-mm lens. The flying height was 970 m (average for the two photos) and the air base was 400 m. The measured photographic coordinates of points A and B in the "flight line" coordinate system are x_a = 45.78 mm, x_b = 127.33 mm, y_a = -32.56 mm, y_b = 57.39 mm, x'_a = -43.96 mm, x'_b = -48.25 mm. Find the length of line AB and the elevations of A and B.

Mini Case Study II: This Is Fine (no it is not) [48]

We know that one of the many indicators of climate change is global oceanic warming, and that our global oceans have been steadily warming as a direct result of anthropogenic-driven climate change. Obviously, anyone who denies this is an idiot (and if you think this, quit scioly immediately!). However, this time we'd like to know *how* this can possibly be observed.



Source: Copernicus Climate Change Service, 2023

51. [4] We know that, for the most part, sea surface temperatures provide good indications to ocean warming trends and thus can be used to assess climate change. Observe the color-coding of the map.
    a. [2] What do the red areas of the map represent? Relative to what?
    b. [2] What do the blue areas of the map represent? Relative to what?

52. [3] Note how there's not a lot of blue on the map relative to the rest of the world. However, the areas that have "blued" over these decades are worth noting. Propose a specific hypothesis as to how and why these areas have turned blue, and the significance of such an event.
53. [4] Although ocean cooling can be considered "good" in the context of climate change, it's important to focus on these outliers too as a result of and indication of climate change. There's a lot of talk of average temperature increases with regards to global temperature changes, especially this "1.5 ℃ global increase".
    a. [2] Contextualize and explain what this means, in regards to global warming.
    b. [2] Although a seemingly small figure, why does a 1.5 ℃ global increase mean so much in regards to the Earth as a whole? Give an explanation to the best of your ability.

There's these two funny laws you'll see popping up when you look into the principles that thermal remote sensing data operates on. Those are the Stefan-Boltzmann law and Wien's displacement law. Since thermodynamics is scary and the writer has not passed multivariable calculus (send help) or reached the level of comprehending quantum physics, we will focus on Stefan-Boltzmann and its relation to the aforementioned case.

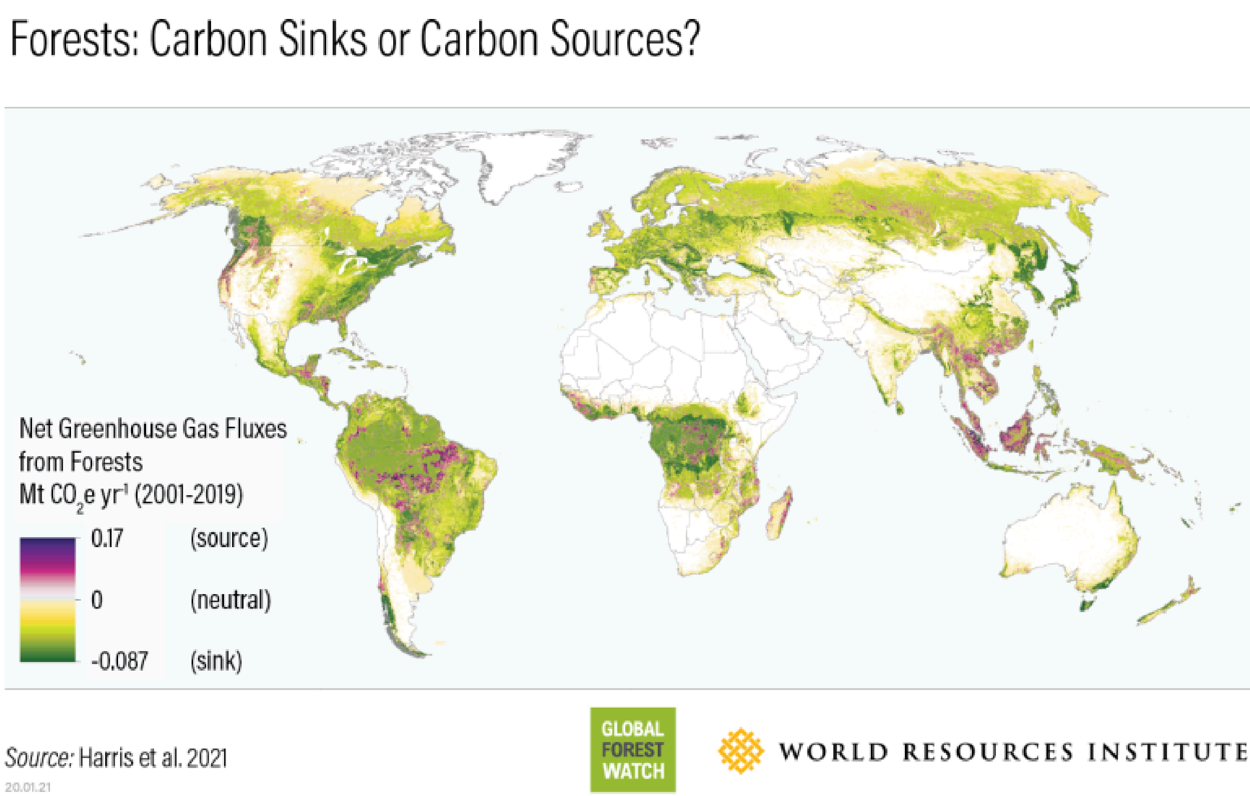
54. [3] In your words, what does Stefan-Boltzmann's law state?
55. [1] What wavelength is thermal data (temperature data specifically) found/measured on?
56. [1] Select following the remote sensing instrument which is best suited to collect temperature data.
    a. LiDAR
    b. MSS
    c. SAR
    d. GPS
    e. TM
57. [2] We know that Stefan-Boltzmann's law is derived with an ideal blackbody in mind. Define a blackbody.
58. [8] In this case, we will do a very oversimplified, boiled down derivation of Stefan-Boltzmann from Planck's Law. Planck's Law, given in this case, is expressed as $(2hv^3)/[c^2(e^{hv/kBT}-1)]$, where h is the Planck constant, c is the speed of light in a given medium, and kB is the Boltzmann constant.
    a. [3] If you can, please give your definition of Planck's Law. How does it apply in this case?
    b. [5] Since we know that power P is equivalent to radial exitance M multiplied over the object surface area A, **derive and set up** an integral that uses the given expression of Planck's Law to arrive at the Stefan-Boltzmann law. **YOU WILL NOT NEED TO ACTUALLY INTEGRATE.** You will not need to simplify to the most commonly found expression of the latter, but it is ideal. Please show as much work as you can (so long as the supervisors can follow your train of thought). Write and set the integrals up following the formatting of **(whatever constants) * [(int from lower -> upper bound) (actual integrand) (variable of integration)]**. Whatever special symbols you use can either be copy-pasted or written out. The integral table is attached as a link below if needed. You may open the file and exit out of the browser without notifying supervisors that you are doing and use it as you see fit. **WE WILL NOT BE GRADING FOR EVALUATION OF THE INTEGRAL.**
       [Integral table](https://www.integral-table.com/downloads/single-page-integral-table.pdf): https://www.integral-table.com/downloads/single-page-integral-table.pdf
59. [2] Obviously, we know that Earth and its components (including the oceans!) are not ideal blackbodies. However, we would like to see how close it can get to such and thus justify the applicability of the Stefan-Boltzmann law to Earth's oceans specifically. Oceans are not perfectly smooth, constant-temperature, non-changing surfaces. Therefore, their emissivity fluctuates over time and location (amongst many other things). What is emissivity? Define in your own words.
60. [3] What is the emissivity value of a blackbody? How does this compare to an average, normalized value of the ocean surface?

61. [2] Therefore, can we conclude that the Stefan-Boltzmann law is applicable to the Earth's oceans? Why?
62. [15] Using Stefan-Boltzmann, propose an experimental process that determines sea surface temperature (SST) via remote sensing instrumentation and applies the aforementioned law.

Mini Case Study III: U u wa wa uwa [54]

Unfortunately, Gerald (remember him?) has succumbed to The Urges and gone full beaver mode 😱. He intends to singlehandedly eat all forest cover until the only plant cover remaining is whatever current vegetation that's being used for agricultural purposes (bro thinks he's Usagi 💔). Unable to convince Gerald to hop off those reels of Usagi sigma edits, you begrudgingly listen to his yapping and hope he develops hypervitaminosis and botulism Type E before he even thinks of attempting another one of his nefarious plans. Dude's basically gone entirely supervillain now. It is TRAGIC seeing how he went from the writer's (Amy's) adopted pusheen to this final boss of ecoterrorism.

63. [2] Firstly, we need to get some definitions out of the way. What is standing biomass, in your words?
64. [3] Do measurements of standing biomass tend to over or underestimate the actual amount of carbon being sequestered in a system currently? Why?
65. [3] We can measure the rates of carbon cycling within a specific area of land or ecosystem, and then see how this can be used to estimate standing biomass cover and thus other measurements such as productivity.



Harris et al. 2021 / Global Forest Watch / World Resources Institute

   a. [1] What type of remote sensing instrument was used to capture this sort of data?
   b. [2] How do those instruments specifically capture the data related to carbon emissions?

66. [2] Note the areas in purple, which are determined to be carbon sources. What is a carbon source? A carbon sink?
67. [4] Why are those aforementioned areas acting as carbon sources? (Think of how humans play a role w/ direct anthropogenic activities that are contributing to this).
68. [2] How does this contribute to climate change?

69. [20] LiDAR, as previously shown earlier on in the test, can be used in many applications that require knowledge of topography and elevation.   Propose a process in which LiDAR can be used to measure standing biomass over a forested stand. (Hint: forests, like other terrestrial features, are 3D. A stand can be envisioned as a "slice" or strip of forest that is typically uniform in terms of something such as composition, structure, age, etc. throughout that can sometimes be generalized to the rest of the forested area).
70. [18] NDVI can also be used to measure biomass, albeit more indirectly. Both NDVI and the aforementioned LiDAR method you described previously, staggered over time, are good for analyzing forest cover loss and degradation.
    a. [6] However, both have their drawbacks. Propose an argument against using the staggered NDVI readings method for measuring standing biomass loss.
    b. [6] Propose an argument against using the staggered LiDAR readings method for measuring standing biomass loss.
    c. [6] If you were to pick one out of the two for this specific purpose, which one would you use, and why?